Suppose the fluid velocity is defined by the baryon current, as in Liam's note,

$$J_B^\mu = n_B U^\mu. \tag{1}$$

This corresponds to choosing the baryon-current (Eckart) frame. In this frame there is no baryon diffusion relative to U^μ, while energy may flow relative to the baryon flow. The version of stress-energy tensor in Liam's note neglects viscous stresses, so

$$T^{\mu\nu} = (\epsilon + P)U^\mu U^\nu + P g^{\mu\nu} + U^\mu q_{\rm th}^\nu + q_{\rm th}^\mu U^\nu. \tag{2}$$

Here heat-flux four-vector is orthogonal to the fluid velocity $U_\mu q_{\rm th}^\mu = 0$. For the metric convention $(+,-,-,-)$ as in our arXiv:2302.09596, define

$$\Delta^{\mu\nu} = g^{\mu\nu} - U^\mu U^\nu, \tag{3}$$

and the four-acceleration

$$a^\mu = U^\nu \nabla_\nu U^\mu. \tag{4}$$

The first-order relativistic Fourier law is then

$$q_{\rm th}^\mu = -\kappa \left(\Delta^{\mu\nu}\nabla_\nu T + T a^\mu\right). \tag{5}$$

Using the ideal-fluid Euler equation,

$$(\epsilon + P)a^\mu = -\Delta^{\mu\nu}\nabla_\nu P, \tag{6}$$

this can be rewritten as

$$q_{\rm th}^\mu = -\kappa\, \Delta^{\mu\nu}\left(\nabla_\nu T - \frac{T}{\epsilon + P}\nabla_\nu P\right). \tag{7}$$

In the local fluid rest frame,

$$q_{\rm th}^i = -\kappa\left(\partial_i T - \frac{T}{\epsilon + P}\partial_i P\right). \tag{8}$$

The usual nonrelativistic Fourier form,

$$q_{\rm th}^i \simeq -\kappa\, \partial_i T, \tag{9}$$

is recovered when the acceleration/pressure-gradient contribution can be neglected.

The second term is an intrinsically relativistic correction to Fourier's law. Thus, a vanishing spatial temperature gradient does not in general imply vanishing heat flux, since acceleration (or, equivalently, a pressure gradient) can also drive the heat current. Conversely, the condition $q_{\rm th}^\mu = 0$ implies

$$\Delta^{\mu\nu}\nabla_\nu T + T a^\mu = 0, \tag{10}$$

or, using the Euler equation,

$$\Delta^{\mu\nu}\left(\nabla_\nu T - \frac{T}{\epsilon + P}\nabla_\nu P\right) = 0. \tag{11}$$

This is the condition of vanishing conductive heat flux; it should not by itself be identified with the adiabatic limit. For dynamical applications, the adiabatic and isothermal regimes are more naturally distinguished by comparing the thermal-diffusion timescale with the characteristic hydrodynamic timescale. Schematically,

$$\tau_{\rm th} \sim \frac{c_V L^2}{\kappa}, \tag{12}$$

up to thermodynamic factors. Then

$$\omega\tau_{\rm th} \gg 1 \quad \rightarrow \quad \text{adiabatic response,} \tag{13}$$

$$\omega\tau_{\rm th} \ll 1 \quad \rightarrow \quad \text{rapid thermal equilibration / isothermal response} \tag{14}$$

Finally, remember that the first-order Fourier law is parabolic and therefore acausal as an evolution equation. If $q^\mu_{\rm th}$ should be evolved in time in your problem, a causal second-order formulation should be used, schematically,

$$\tau_q\, \Delta^\mu{}_\nu D q^\nu_{\rm th} + q^\mu_{\rm th} = -\kappa\, (\Delta^{\mu\nu}\nabla_\nu T + T a^\mu) + \cdots\,, \tag{15}$$

where $D = U^\mu \nabla_\mu$ and τ_q is the heat-flux relaxation time.